

Tezuka's Literary Robot:

Instrumentalism, Sentimentalism, and Mech-aesthesis in *Mighty Atom*

Brian Milthorpe

Abstract: This article investigates how the robot literature of Unno Jûza and Karel Čapek establishes the legibility of Osamu Tezuka's most influential robot character: the "child of science" Atom from the manga series *Mighty Atom* (*Astro Boy*). While critics have noted specific allusions to Unno and Čapek in Tezuka's work, this article posits that the influence of these authors exceeds limited sources of reference material. Instead, it argues that through *Mighty Atom* Tezuka performs a strong misreading of Unno's fiction by re-interpreting themes and motifs of technoscientific autonomy sourced from Unno through a form of humanism derived from Čapek's *Rossum's Universal Robots* (1920). The first section discusses how Unno's robot fiction entangles technoscientific autonomy and nationalistic instrumentalism. The second section demonstrates how Tezuka transforms this discourse through reading Tezuka's *Doctor Mars* (1947) and *Mighty Atom* (1952-1968) alongside the Unno novel *Chikyû yôsai* (1941, transl. Earth Fortress). The third section establishes how Čapek furnishes a model for the reorientation of technoscientific autonomy toward the sensitive heroism of Atom. The article concludes by reflecting on how this revision of Unno through Čapek inheres within robot science fiction comics and animation succeeding *Mighty Atom*.

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Osamu Tezuka held the figure of the phoenix in high regard. This symbol of death and rebirth signified not only the lifelong path to his unfinished magnum opus *Phoenix* (1954-1988) but also the constant exchange of victories and setbacks that defined his prolific output in both animation and comics. However, if the phoenix goes some way toward describing the rhythm of Tezuka's life and work, a humbler bird comes closest to symbolically approximating the way in which he structured and populated his narratives. When it came to crafting stories, Tezuka was a magpie. As Natsu Onoda Power notes, Tezuka supplied his stories "by combining vocabularies and techniques from multiple forms of art and other discourses," collecting the shiny odds and ends of cinematic camera techniques, literary plot devices, and theatrical motifs from a voracious lifelong consumption of film, theater, literature, and music.¹ Tezuka himself was often at pains to flaunt his magpie status. Series revisions and interviews frequently enumerated the sources that stocked his imagination of a given scene or character: the afterword of *Metropolis* (1949), for instance, references Lambert Hillyer's *The Invisible Ray* (1936) as the inspiration for the death scene of the artificial being Michi.² If self-conscious citation and adoption were intrinsic to Tezuka's compositional practice, however, flaunting this encyclopedic recall of works raises a thorny embedding problem for analysis: it becomes impossible to differentiate copying from creation.³

An anecdote hitherto unpublished in English provided by Shimamoto Mitsuaki represents this problem in stark detail. In a meeting arranged by Takumi Shibano, editor of the SF magazine *Uchūjin*, Shimamoto joined five or six other teenaged science fiction fans for an interview with Tezuka to be published in *COM* magazine. Whether out of anxiety or respect, Shimamoto asked Tezuka a softball question regarding the obvious

influence of the Unno Jûza novel *Taiheiyô majô* (1939, transl. Pacific Demon Castle) on the *Mighty Atom* chapter “Sea Serpent Island” (1953). An engineer at the Ministry of Communications Electrical Testing Laboratory researching wireless and vacuum tube technologies, Unno Jûza (the penname of Sano Shôichi, 1897-1947) began his writing career with the 1928 publication of the detective story “Case of the Strange Death in the Electric Bath.” He followed this debut with a rapid series of air raid stories that imagined the violent consequences of an anti-scientific national ideology, an output that garnered the critical attention of the Japanese military.⁴ Throughout the growing militarism of the 1930s, Unno continued working in detective and adventure fiction, tingeing the mechanics of these genres with a style imbricating escalatory pulp, jingoistic heroism, and erotic-grotesquery with the technologies of radio and television, the biological and physical sciences, and the noir aesthetics of spy espionage. So in asking about Unno, Shimamoto only drew on the widely reported fact that Tezuka had avidly read *Pacific Demon Castle* along with other Unno novels like *Chikyû yôsai* (1940-1941, transl. Earth Fortress) and *Kasei heidan* (1939-1940, transl. Mars Army Corps) during his youth.⁵

However, Tezuka’s answer flew in the face of this common knowledge: “I haven’t really read as much Unno as people say. And since I’ve never read *Pacific Demon Castle*, it’s highly unlikely that I received any influence from it.”⁶ Reflecting on why Tezuka denied the influence of Unno against all evidence to the contrary, Shimamoto offers the hypothesis of an anxiety of influence: “Unno’s ideas and [Tezuka’s] thinking were so integrated, it wasn’t clear to the artist himself where his original ideas came from... For Tezuka, Unno was not only a respected veteran, he was a rival to be persistently

overcome.”⁷ What Shimamoto detects in *Mighty Atom* and what Tezuka pretends not to hear is that voice Harold Bloom discerns where he writes, “the dead may or may not return, but their voice comes alive, paradoxically never by mere imitation, but in the agonistic misprision performed upon powerful forerunners by only the most gifted of their successors.”⁸ In light of reports that Unno pinned hopes on Tezuka to act as his heir in the realm of sci-fi before his death in 1949, such an obtrusive denial in front of the next generation of fans provides Tezuka an opportunity to stress his credentials as Unno’s successor, rather than constituting outright subterfuge or momentary mnemonic lapse. He presents his singular “misprision” of Unno in *Mighty Atom* as a new, unblemished paradigm for creativity.

Further credentialization along these lines, both implicit and explicit, has not gone unnoticed by commentators.⁹ Inoue Haruki, for example, points to Tezuka’s blatant appropriation of mechanical designs from Kikuzô Itô’s illustrations for Unno’s *Mars Army Corps*, *Earth Fortress* and the short story “Jinzô ningen no himitsu” (1940, transl. “The Secret of the Android”).¹⁰ Inoue’s critical eye sees Tezuka lifting the angle of rocket launches, the eccentricities of coiled mind-control helmets, and the regularity of regiments of gun-wielding robots directly from Kikuzô into *Ambassador Atom* as well as the “Mission to Mars” (1953) and “The Robot Spaceship” (1963) chapters of *Mighty Atom*.¹¹ For Inoue, these references expose Tezuka’s “artisan” status by furnishing proof of the absorption and recreation of the mechanics of previous masters of both visual and literary fiction in the wake of Japan’s 1930s “Robot Boom”--- a cultural phenomenon undergirded by the Japanese publication of Karel Čapek’s *Rossum Universal Robots* in 1923 and sparked with the 1929 invention and exhibition of Japan’s first modern robot,

Nishimura Makoto's *Gakutensoku*.¹² Beyond this identification and categorization of visual and generic quotation, however, no study to date has focused on the sphere of anxious influence that Shimamoto's anecdote raises: namely, the question of how Unno's textual voice within its robotic context shapes Tezuka's work *despite* conscious adoption of plots or motifs.

This article works to supply the provisional framework and content for such investigation, specifically where it concerns the provenance of Tezuka's most iconic character: Atom from the science fiction series *Mighty Atom*, featuring a state-of-the-art atomic android driven by a ten-thousand-horsepower nuclear engine created by Dr. Tenma at the Department of Precision Machinery of the Ministry of Science after the sudden death of his son Tobio. The serial format and science fiction superhero genre of *Mighty Atom* dictates that Atom perform the duties of a regular schoolboy while also solving crimes, repelling alien invasions, and decommissioning through physical force errant robots and evildoers. A matrix of motifs drawn across generic boundaries establishes Atom's image: the speed lines of his jet rocket feet and outstretched fists in flight cite the intrepidity of *Superman* (1938); the glossy spikes of Atom's jet-black hair set up an endearing visual rhyme with the ears of *Micky Mouse* (1928); and the sympathetic circumstances of his early life owe to Walt Disney's *Pinocchio* (1940). While these frequently cited elements detail Atom's actions, appearance, and origins, they say relatively little about his ontology or ethics – what he is and why he does what he does.

Answers to these questions can be gleaned from another, less specific snippet of autobiographical comment again gravitating around the literary environment of Japan's 1930s Robot Boom. In his *Astro Boy Essays*, Frederik L. Schodt reports that "as a youth

the robot stories that influenced [Tezuka] most were a translated copy of *R.U.R* he read in 1938 and the prewar works of a Japanese science fiction novelist named Jûza Unno...”¹³ Following Schodt’s lead and Shimamoto’s suggestion, the sections below track how the prewar robot literature of these two authors makes the sentimental heroism of Atom not only legible but also possible. In Unno’s detective and science adventure fiction, individual and national fortunes often depend on rational confrontation with the raw elemental power of scientific technologies. Structuring this heroism is a correspondent “scientific spirit,” the self-instrumentalizing attitude of the technician capable of uncovering, constraining, or directing the autonomous forces of technoscience. Far from rejecting this discourse, Tezuka literally embodies it through the figure of an individualized autonomous boy robot. *Mighty Atom*, however, redirects tropes of technical competence away from national duty toward a practice of sentiment evinced in Čapek’s *R.U.R*. The result is Atom himself: a sentimental instrument, a caring and careful weapon whose isolated emotion offers a revised scientific spirit based on negotiating with the discourses of autonomy and automatism circulating within early twentieth-century robot fiction.

Independent Instruments - Unno Jûza’s Robot Fiction

At first glance, Atom exists nowhere near Unno. This is because the predominant tendency of Unno’s writing pictures robots as remorseless weapons or mindless instruments, whether in the form of advanced military technology or egomaniacal invention. In the words of Sena Takahiko, “most robots that appear in these works are not the elaborate creations of modern science fiction films with electronic brains capable of human behavior. They are clunky human machines who work according to human

command.”¹⁴ Yet it was precisely the technical plausibility of Unno’s robots that in part defined the style of *kagaku shôsetsu*, or the proto-science fiction novel around which his reputation cohered. Through stories buttressed by historical awareness and populated with “well-informed references to the hard sciences,” Unno’s work addressed “the transformation of everyday life by new forms of scientific knowledge and technological prosthesis” during the early Shôwa period.¹⁵ This technical and historical grounding shaped early modern Japanese robot literature into a “site where debates surrounding this mechanization of the human body were discussed and contested” through dramatizations of the moral degeneration, political intrigue, and aesthetic excess associated with a rationalizing modernity.¹⁶

Unno demonstrates his attention to the historical repertoire of command-and-control robots in an early essay titled “Artificial Tales” published in the April 1930 issue of *Shinseinen*, a journal pivotal in the development of Japanese science fiction through its cultivation of the *ero guro nansensu* (erotic grotesque nonsense) aesthetic: a “pre-war, bourgeois cultural phenomenon” engaging the “deviant, the bizarre, and the ridiculous.”¹⁷ His survey spans from the Westinghouse Corporation’s Televox to recent public spectacles and events, such as an exhibition “at the Tokyo-Ueno Matsuzakaya department store where an android took the stage and fraternized with patrons” and reports of a robot dog created out of telephone equipment, motors, and electromagnets.¹⁸ Cited among these real-life robots is the “first ever made... who waddled around London and startled its residents. He was able to sit, stand, and wave his hand. He somewhat resembled a human tank, or the reincarnation of a Roman soldier in full armor.”¹⁹

Although unnamed, Unno is likely referencing Eric, a motorized robot exhibited at London's Royal Horticultural Hall in 1928 by his inventors Richard William and Allen Reffell that went on to influence robot design in Japan. With his polished metallic body, angular features, and crown-shaped head, Eric resembled a suit of armor able to sit, stand, and speak with the aid of a radio receiver (Figure 1). In the short story “Jinzôningen jiken” (1936, transl. “The Android Incident”), Unno casts a robot with a distinct resemblance to Eric as a violent murder weapon powered by radio technology. While meandering the streets of Tokyo one autumn evening, Homura Sôroku---a gifted physician and amateur detective whose name puns on Sherlock Holmes---stumbles his way into investigating the gruesome death of Prof. Takeda, a nationally renowned inventor of electromechanical androids bludgeoned to death in his sleep. The sole witness to the murder is Model Q No. 8, a robot “stored in a box leaning against the wall like a coffin...” described as “a medieval knight ... with a protruding jaw and long nose jutting out from his face above a deep-cut crescent aperture within which one could see a speaker...”²⁰ A logical trail leads part-to-part from Model Q No. 8's operation manual, to a radio placed in Prof. Takeda's laboratory, and, finally, to an instruction set of violent movements pre-programmed into the robot by the professor's nephew and activated by codewords integrated into the script of a military-themed radio drama. The mechanics of the detective novel thwart mechanical murder.

While many of Unno's robots emulated the pattern of Model Q No. 8 in amounting to complex remote-operated electro-mechanical puppets, “The Android Incident” also subtly reveals how Unno's conception of the artificial human was also alloyed toward more of a plastic, premodern understanding. Early on, Homura chides

his colleagues as they stand dumbfounded before the robot “which did not stir an inch, and yet appeared as if it could spring bellowing from its box at any moment,” saying “there’s no way you could’ve concluded that a real soul is inhabiting this steel machine, is there?”²¹ The narratorial apparatus also uses tropes of technical complexity and functional equivalence to tease the line separating human from machine when the radio broadcast stirs Model Q No. 8 into motion: “Metallic clanks and clangs were emitted as the android took a step from its box – as though a spirit inhered within it.”²² Even as Homura is convinced nothing supernatural is occurring, the notion of living independent machines presents itself as a constant backdrop.

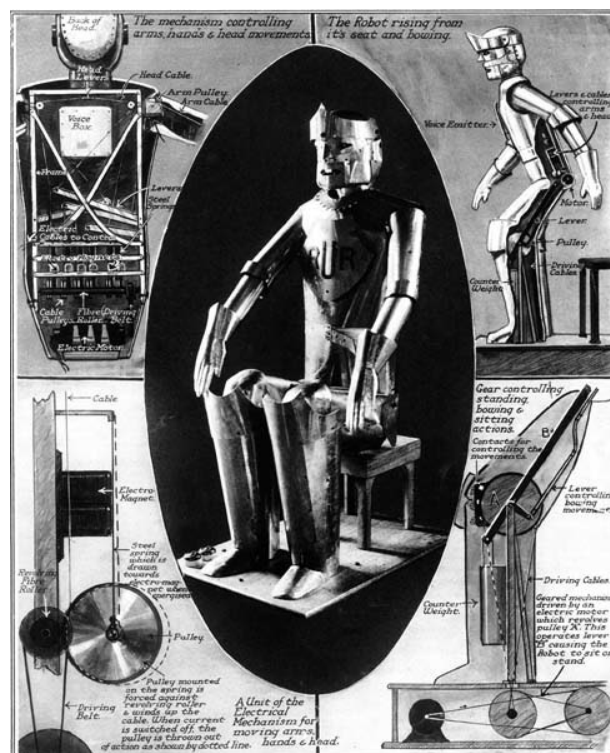


Figure 1: A diagrammatic advertisement for Eric the mechanical man, demonstrated at the Royal Agricultural Hall, Gomshall, England 1928. Courtesy of Photofest Digital.

Unno's "Artificial Tales" essay sheds light on the justification for this background.

Before introducing real-world units like Eric, Unno intimates that the reproduction of

the human form evinced in these modern inventions recalls a pre-modern mythic context. Examples drawn from this context range from the construction of Eve from the rib of Adam, the birth of Japanese gods in the *Kojiki*, and the autogenic ability of the Monkey King Sun Wukong from the 16th century Chinese novel *Journey to the West*. At the end his essay, Unno reintroduces premodern notions of elemental proliferation and transformation through a distinctly modern (and macabre) image:

... we can't rule out the possibility of a world war occurring before long in which scientific warfare will wipe out all biological humans...After this, there will only be the continued activity of soulless machines. When I imagine such a desolate picture, my brain shrivels up and I feel as if I'll go mad...²³

Here Unno anticipates that aspect of his fiction “delighting in the havoc that the excesses of science... wrought on traditional measures of identity.”²⁴ But Unno's essay shows how this delight in the transformative power of modern science is not just forward-facing: it begins in proper with premodern fables and legends. The playful sentiment expressed here approximates art historian Itagaki Takao writing in 1929 that “Whereas in former times, people believed in religion, the people of today believe in the power of machines... As today's social lifestyle idealizes ‘the rationalization of life,’ new religion is born from the authority of machines.”²⁵ In this sense, the image of apocalypse that concludes Unno's essay provides the corollary of Homura's enlightened attitude. Both raise before young readers the possibility of a contest between minds acting like machines and machines adopting minds. In the end, the same transcultural and transhistorical will to reproduction that produces the robot may, through rational

technoscience, exceed its producer and attain a self-sustaining autonomous perfectibility.

Both a satire on Japan's imperial ambitions and a paean to its scientific endeavors, Unno's "Eighteen O'clock Music Bath" (1937) narrativizes this conjunction between premodern technical power and futuristic technoscience through picturing the proliferation and eventual independence of artificial humans. This short story imagines an authoritarian society where citizens are subjected to the indoctrination of "music baths," a technology invented by the "star scientist" Professor Kohak through which national anthems stimulate neuronal activity and boost economic productivity. After a mysterious explosion obliterates Kohak, President Miluki and the Secretary of State Madam Asari inspect his laboratory and find there a "museum of automata... beginning with items from the 4th century BC" that materializes the long line of robot history Unno outlines in his essay: "marionette-like contraptions and armored warriors, progressing to radio-controlled units built with relays, and remarkably human-like models with artificial skin."²⁶ Prime among these is a prototype gynoid "reminiscent of Venus de Milo" named Annette.²⁷ Jealous of the sexualized response this gynoid provokes, Madam Asari surreptitiously feeds her synthetic flesh to the president and multiplies the frequency of music baths to twelve times a day, asserting that "science can never dominate government, but government will invariably dominate science."²⁸ In the final hour, however, Kohak miraculously returns in a "strange beetle-like suit of armor" leading a procession of "roughly 500 androids resembling Annette" to seize control of a military decimated by excess patriotism and institute a self-legitimizing android nation.²⁹

Written during the Soviet-Japanese border conflicts, the novella “Jinzô ningen efu shi” (1939, transl. “Android Mr. F”) recapitulates this valorization of robots as technological products whose inherent rationality outstrips the shortsighted contests of political ideology. It opens on the scene of the Russian scientist Professor Ivanov welcoming a young boy named Shotaro and his sister Mariko to an annual exhibition of animal automatons. Ivanov entices Shotaro into his basement laboratory with promises of witnessing his latest creation: an android capable of speech. Imprinting the likeness of Shotaro onto this mechanical boy through some sort of electrical field, Ivanov uses the flying, digging, and mind-reading doppelgänger Mr. F to wreak havoc on the military by sinking ships and melting tanks.³⁰ After Detective Homura is enlisted to help Shotaro clear his name, the pair destroy the remote-control unit operating Mr. F only to discover that Ivanov was himself an android from the very beginning. The novella ends with an unsettled reflection from Homura: “What a terrifying degree of scientific power this is. We Japanese have no choice but to pull ourselves together!”³¹ Condensed in this final line is the predominant message of Unno’s robot fiction during Japan’s growing nationalism: the youth of imperial Japan must contend with the instrumentalization of an elemental technoscience whose tendency toward violence and transformation both enlivens and threatens the stability of nations and identities.

It was the complexity of this message that allowed Unno’s fiction to evade the harsh regime of government censorship promulgated in the late 1930s. By framing technoscience as an autonomous force that must be wrangled and developed by national interests, Unno’s “repeated warnings to cultivate scientific power... in order not only to win wars but also sustain peace” rang utterly ambiguous.³² They comported with the

purpose of propaganda only to divert attention away from the nation. Readers learn to “pull themselves together” within and for a transhistorical force that outstrips intermittent drives for national consolidation. In both form and content, Unno’s fiction “beguiles his readers with extreme yet conceivable plotlines so that their interest in science and its possibilities will continue long after they leave the story.”³³ Beguiled in this very way, Tezuka effectively heeded “Unno’s call for scientific mobilization by becoming an author of manga informed by science.”³⁴ But this science-informed manga would remain in touch with Unno and develop a method for communicating the spirit of scientific mobilization underwriting his fiction. It would shed its nationalist overtones and preserve the semiotic network between the autonomous, the outré, and the heroic. With this, it becomes possible to say, if only in a reserved sense, that Tezuka found blueprints for Atom, along with his friends and foes, within and between the thousands of lines of Unno’s *kagaku shôsetsu*.

Building the Child of Science with *Earth Fortress*

In contrast to Unno’s presentation of robot and android figures like Model Q, Mr. F, and Annette as speculative augmentations of pre-existing technoscientific processes, early twentieth-century Japanese popular culture traded between the more eccentric and mundane dimensions of the artificial humanoid. Gajô Sakamoto’s *Tank Tankuro* (1932), for instance, conveyed the mobilizing tendency of military science through its protagonist, a top-knotted rambunctious soldier capable of deploying propellers and machine guns from his porous iron body and of self-replicating to create an army of mini-tanks. In contrast to this whimsy, Ôshiro Noboru hyperbolized the basic visual elements of gears, bolts, tubes, and rivets to give otherwise imposing robots a

welcoming disposition in manga such as *Yukai na tekkôjo* (1941, transl. The Joyful Ironworks), a work that self-consciously depicted the imaginative industrial colonization of Manchuria. The Japanese toy industry reinforced the spread of these mechanical design elements through wind-up tin plate toys such as the pre-war Robot Lilliput (c.1937) and post-war Atomic Robot Man (1949), further ingraining the blocky robot motif in young minds (Figure 2).



Figure 2: Robot Lilliput N.P 5357, a wind-up tin-plate toy manufactured in Japan in 1937 considered to be one of the first mass-produced robot toys. Courtesy of Vectis Auctions Ltd.

While familiar with the previous generation of manga robots, especially the work of Ôshiro, and steeped through his middle-class upbringing in the techno-cultural milieu where automatic toys from ambling robots to film projectors circulated, Tezuka stresses in a 1978 interview the degree to which Unno disrupted his generation's attachment to these whimsical and clunky synthetic humanoids:

The image of robots we had when we were kids was one where their necks, torsos, legs, and arms were wired together. All the robots in manga and book illustrations were that way. Sort of like spring clockwork, being able to stretch and shrink freely [...] The robots back then didn't really act like humans, or, I should say, the impression of them was no more than a toy kicking into action. But wasn't it the sort of thing like *Earth Fortress* that really opened our eyes to the depiction of a proper robot?³⁵

Serialized in the magazine *Shônen Shôjo Tankai* between August 1940 to February 1941, *Earth Fortress* surged with the kinds of militaristic fodder designed to rouse the sentiment of young male readers during the Pacific War, including wonder weapons, submarine battles, and space-time dilation, effectively “conflating lived experience, fantasy, and propaganda.”³⁶ The novel concentrates on the heroic exploits of Prof. Kurouma, a government-military agent who labors to uphold his duty to defend Japan using his island-sized nuclear fission submarine, Kurokuro Island. After intercepting a strange Venusian radio transmission from the future, Kurouma navigates rising tensions for natural resources between militarized power blocs and the taunting interference of a mysterious transdimensional being known as Ambassador X.

Vital to these exploits is Kurouma's beautiful robot companion Princess Olga, who serves a dual role as sidekick and technical instrument. Unno establishes her exceptionality early on by picturing her deftly manipulating the blinking switches of complicated control panels and precisely relaying information from Kurokuro Island's guidance system (although not without a kind of macabre humor when, after an attack

by Ambassador X, her head is seen rolling besides Kurouma's feet). Upon revealing her machinic identity through this scene of decapitation, Kurouma praises this artificial human for being "exceedingly suited to the role of an assistant... She unflinchingly follows orders, has no reason for food or laziness, and, if the need to exchange parts arises, can be used without restraint."³⁷ This sober utility is disarmingly combined with "a beauty also rare in the human world;" indeed, Kurouma "toured the famous art museum in Milan for a week, at long last selecting her model" by choosing his "ideal picture of femininity" from amongst historied paintings and sculptures.³⁸ With resonances of Villier de l'Isle-Adam's *Tommorow's Eve* (1886), published in Japan in 1937, Unno thus supplements his wartime heroics with a disconcertingly ambiguous gynoid, both instrument and lover: an artificial being who can be exchanged for mechanical parts and who instills in Kurouma, whenever he sees her, "a sense of assurance similar to that felt while being held in [his] mother's embrace."³⁹

This tension between pure instrumentalism and desire-laden aestheticism sets up a pattern re-deployed throughout the novel. After the disappearance of Kurokuro Island, Kurouma enters one of a handful of soliloquies conveying his disappointment and isolation: "I never missed a living, breathing human being more than at this moment. Princess Olga, now at my side, would faithfully follow my every command, but, unfortunately, as an android, could never address or consult with me."⁴⁰ Midway through his exploits, Kurouma delivers a didactic justification for this tension between succor and duty in the form of Unno's exhortation to wrangle the alluring autonomy of technoscientific forces: "As science progresses, humans have no choice but to endure lives of solitude. For a single human must operate a great many of machines...."

Otherwise, we will fail to manifest enough scientific power to curb the invasion of foreign adversaries.”⁴¹ In this light, Kurouma’s monologues serve to remind young readers that Olga is a presence that both assuages and instantiates the inevitable isolation of modern science and technology. As a figure combining ideal beauty and ruthless technical efficiency, to become familiar with her is to prepare for the future. The conclusion of *Earth Fortress* recapitulates this lesson in bold outline. Rather than a grand final battle, it is the development of Japan’s technoscientific power in the form of a fourth-dimensional oscillation repulsion device and giant concrete shell encapsulating the island that calms hostilities by convincing the Venusian Ambassador X to cease his provocations and leave room for Earth to develop its technology.

Olga’s independent spirit, superior technical functionality, and attractive form, tinged with Kurouma’s sense of loneliness and isolation, exerted a strong influence on Tezuka, even enough to cast the android in his 1980 science fiction film *Phoenix 2772*. In the same interview where Tezuka stresses the importance of Unno in shaping a technologically grounded conception of robots, reminiscences of Olga make this lasting influence explicit:

... there was a helper robot who looked after the main character. She was an android that had the ability to think and could answer any question asked of her just like that – a really remarkable robot... ah! I remember now. Princess Olga... You remember that don’t you? I was enamored with that, you know. I thought it was so wonderful, I put it in a manga, *Doctor Mars*, I think.⁴²

Published in paperback in 1947, *Doctor Mars* illustrates the exploits of a young boy named Kenichi liberating his grandfather Dr. Bouton from Jellyfish Island, a robot production facility run by a Martian disguised as an Earthling named Dr. Poppo who develops a weather modification device to conquer Earth. Hoping to convince Dr. Bouton to aid in his research, Dr. Poppo introduces two classes of artificial humans produced on his island, Type A (clunky robot laborers) and Type B (beautiful android servants). Pîko, an intelligent Type B robot who aids in calculations, swoops in to save Kenichi from execution by Dr. Poppo's henchmen and learns about the nefarious actions of her creator. Instead of force, Pîko peacefully confronts Dr. Poppo with the immorality of his actions. After the failure of this discourse to effect any change, Pîko commands the Type A robots to find Kenichi and "show no mercy" in suppressing any interference from Dr. Poppo's henchman, responding to accusations of treachery by saying "It's because you're a bad person! No robot wants to associate with a bad person!"⁴³

Takeuchi Osamu notes how Pîko's actions in this scene instantiate an early instance of Tezuka putting into practice the equation "high grade robots = possessors of innocence," a pattern "inherited by the *Atom* series and expanded on by the robots it would introduce."⁴⁴ Yet Pîko also propagates the attributes attached to the model of the robot figure presented in Olga and throughout Unno's fiction. Indeed, Pîko's innocence depends more fundamentally on an unspoken assertion of independence, itself supported by intimidating technical capacity (in this case, her ability to fly, her metal body, and her super strength). Where Pîko reasons that no self-respecting robot would knowingly serve an evil master, her very ability to speak out expresses the intertwined positions of Olga in relation to Kurouma in *Earth Fortress*, at once technical instrument

and technician: she stands alone, unperturbed, and efficient, instilling in the reader through her rational strength the same sense of reassurance and security that Olga provided for Kurouma. In the final chapter of *Dr. Mars*, Pîko displays this capacity in a full-page action panel compositionally homologous to an earlier scene depicting the Type A robots scattering Dr. Poppo's henchman.⁴⁵ Like Olga, Pîko demonstrates the degree to which innocence and independence are always buttressed by a potentially violent technical efficiency. And it is this latent efficiency that defines Tezuka's robots by introducing a dash of ambiguity into the otherwise firm distinction between the remote-controlled robot and the autonomous self-operating hi-tech android.

Tezuka's emendation of the figure of the robot received from Unno thus obtains his development of an artificial human whose potential independence both embodies the autonomy of technoscience while also introducing doubts about the consequences of that autonomy. Numerous commentators have noticed this same contradiction operating in different forms across Tezuka's work. Takeuchi, for instance, locates the founding tension of Tezuka's output in a deeply embedded "trust toward science and distrust toward a co-existing scientific culture."⁴⁶ Witnessing the devastation wrought by Japan's mobilization of science and technology during the Pacific war, Tezuka, himself "a recipient of scientific education," comes to mistrust the rationalistic narratives of a collective scientific culture while still "granting life" to the various robots, aliens, and neo-humans of his manga through rationalistic explanations derived from science.⁴⁷ Iwamoto Yûichirô points to a related contradiction, arguing that Tezuka's robot is both ultra-functional and yet deeply limited: "a kind heart and superhuman powers represent the reader's imagination of the universality of science" but "beyond

this representation, what is important is that this universality of science is under human control.”⁴⁸ Both Takeuchi and Iwamoto recognize a tension presented to Tezuka in ready-made form throughout Unno’s fiction by robots and androids such as Mr. F and Olga: these artificial humans are containers for the independence, efficiency, and autonomy of the same technoscience that also makes them into discrete instruments of violence and isolation.

Like his forebear Piko, an apparatus for Tezuka’s trust in an independent and self-justifying technoscience, Atom epitomizes the struggle against the promises and limitations of his own technical autonomy. He is, in other words, a descendent of Olga and Kurouma in embodying an archetype of the “lives of solitude” predicted in *Earth Fortress*, an artificial human who must live a life defined by artificiality in order to ward off external threats of conflict and internal threats of his own instrumentalism. The circumstances of Atom’s first appearance as a side character in the series *Ambassador Atom* (1951-1952) are keyed directly to this identity in picturing the interruption and continuance by other means of a modern life lived through artifice. Before becoming a robot, Atom was first Tobio, the only son of Dr. Tenma, the head of the Ministry of Science. Tezuka first depicts Tobio training his skillful manipulation of machinery by whizzing around in a hi-tech miniature automobile under the supervision of his father. When Tobio is killed in a traffic collision, it is the interruption of this image of a young boy in command of an apparatus that ultimately drives Dr. Tenma’s to robotize Tobio, effectively arresting the development of this boy and his machines by means of their fusion. Dr. Tenma’s frustration that his robot son cannot physically grow leads him to sell Tobio to the circus. Now isolated as a pure technical spectacle like Eric, the boy-

machine Tobio continues practicing his interrupted technical facility, performing advanced mathematical calculations and building perfectly symmetrical cubes from toy blocks.

In his revision of this backstory for the “Once Upon a Time” series serialized in the *Sankei* newspaper between 1968 and 1969, Tezuka significantly expanded this episode to emphasize both the advanced technical specifications of Tobio’s fated car and the technical knowledge and skill required to possess it as a technological object. Dr. Tenma visits a department store to purchase “a life size plastic model air car, faster than the real thing” and brings it home to Tobio, who immediately begins constructing the model with his tools.⁴⁹ After the collision, Dr. Tenma reasons that insufficient technical ability facilitated his son’s death, saying that “if Tobio had been a robot, he wouldn’t have had an accident. He would have been strong and powerful... that’s the way my son should be.”⁵⁰ From this, Dr. Tenma’s proposal to the board of the Ministry of Science to build a one-hundred-thousand horsepower machine capable of feeling directly replays Tobio’s interrupted technical facility, making him a human-machine whose technical knowledge and skill becomes reflexive: rather than constructing and manipulating toy models, he manipulates his constructed body to resolve the threats of modern technoscience and the crimes of those who seek to instrumentalize it. Alicia Gibon’s observation that the newly-born “Atom faithfully studies the relationship between words and things and earnestly embarks upon the project of becoming Tobio” points to this same reflexivity. Atom must become who he was: not simply a human, but also a young technician in training.⁵¹

This pattern of revision continued in the Asahi Sonorama edition of *Mighty Atom* published in 1975, where Tezuka accentuates the crazed desire of Dr. Tenma not simply to resurrect his son through advanced robotics, but also to restore and complete Tobio's precocious command of machinery. In the laboratory scene of Tobio's rebirth inspired by the creation scene of James Whale's *Frankenstein* (1931) which Tezuka would redeploy in multiple chapters of *Mighty Atom*, Dr. Tenma commands a computer not to bring his son to life, but to "breathe **reason** into this mechanical doll!"⁵² Subsequent panels illustrate where Dr. Tenma wishes to see Tobio's new reason put into action. Before his abandonment, the young robot boy can be seen playing with a robot toy resembling Robot Lilliput or Atomic Robot Man and intricate model planes, manipulating diminutive versions of his own machinic form and capability.⁵³ In Gibson's words, "just as a troubled humanity struggles with a new form of power and its attendant ethical dilemmas, Atom's control over his capacities is not complete: he has the strength of a superhuman, but only a boy's control of these awesome powers."⁵⁴ But the discourse of control gravitating around the circumstances of Tobio's death and Atom's birth do not simply look toward the uncertainty of the post-atomic future. They remain in direct touch with the tropes and discourses of Unno's pre-atomic robot fiction.

The Soul of the Instrument; Or, Why Atom Feels

Before the debut chapter of *Mighty Atom*, Tezuka expressed reservations about having the robot boy from *Ambassador Atom* serve as a main character in his own series. His response to the editorial recommendation for an *Ambassador Atom* spin-off centered the demurral that "robots are cold machines."⁵⁵ It was ultimately Tezuka's decision to

make Atom a “reverse Pinocchio” character that helped ease these misgivings: Atom’s robotic functions would create dilemmas whose resolution required a greater sensitivity to the nuances of human emotionality and sociality. Yet the slightly tortured concept of a “reverse Pinocchio” alone does not explain the affective range or depth Atom exhibits. Nor does it address the interiority invested in lifeless semiotic markers that Eiji Otsuka identifies as the hallmark of Tezuka’s characters: unaddressed is the question of why, in Komatsu Sakyo’s words, this boy robot should be a “crybaby.”⁵⁶

Despite Tezuka’s initial disconcertion, the prospect of a sentimental robot would have already been familiar to him through the conduit of Karol Čapek’s *Rossum’s Universal Robots (R.U.R.)* through translations in 1923 by Suzuki Zentarō and Itsuo Oga. While the allegorical contest between capital and labor powering Čapek’s play marked the genesis of numerous tropes now associated with robot fiction, the surface of *R.U.R.* is preoccupied with the question of how an artificial human might become sensitive. Founded after Young Rossum industrialized his father’s quasi-alchemical experiments to discover the secrets of life, the Rossum factory now meets the labor demands of the world through the production of Robots, a term coined by Čapek from the Czech *robot* meaning “drudgery” or “servitude.” These unemotive and sexually sterile organic laborers are mass-produced in vats, molds, and spinning mills for nerves and veins. A young woman named Helena initially seeks to infiltrate the Rossum company to provoke the Robots to revolt on behalf of the International League of Humanity, an institution that views their brutish treatment as a violation of human rights. Despite verifying this treatment after learning about how Robots exhibiting pain are crushed to death, Helena ends up joining Rossum. From fear of resentment, Helena

requests that Doctor Gall, head of the Psychological and Experimental Department, refine the Robot's "temperament" by turning their limited potential for suffering into the depths of a soul.⁵⁷ This humanitarian gift intended to lighten their burden, however, leads to self-awareness and sparks a militarized revolution.

While the Robots of the play depart from the tottering metallic trunks to whom they would lend their name, they nonetheless extend the critique of how existing technoscientific processes such as mechanization and industrialization distort and deaden pliable human bodies and minds. Underlining this technological critique is an anti-war sentiment based on Čapek's experience of World War I, a conflagration "full of new technologies: tanks, grenades, mortar bombs, machines guns, poison mustard gas, and zeppelins."⁵⁸ Not just metaphors of the massifying power of capitalist production or socialist ideology, the Robots exhibit the deadly force of technological efficiency in their uniformity and number, multiplying like the long belts of identical mass-produced bullets fed into machine guns. Through its vision of humanity-qua-machine, *R.U.R* perpetuated an awareness "of the very real possibility of the destruction of the human race by its own machine-based creations."⁵⁹ Writing in London's *Saturday Review* to intervene in a debate between George Bernard Shaw and G.K. Chesterton about the meaning of *R.U.R.*, Čapek himself emphasized this possibility, writing "those who think to master the industry are themselves mastered by it. Robots must be produced although they are, or *because* they are, a war industry. The conception of the human brain had at last escaped from the control of human hands."⁶⁰ Once exteriorized into a logical and self-operating system, the brain and hands see the dominion of their autonomy ceded to automatism.

The first half of Čapek's play thus prepared the way for Unno's preoccupation with robots as self-legitimizing instruments of war. Although both authors concur in their modernist vision of the “rational, autonomous human being as an object of conditioning and control,” they depart in their conception of the outcomes of this conditioning.⁶¹ For Unno, the struggle for autonomy against automatism creates space for a heroism which overlaps scientific thinking and the efficient function of state machinery. In the final act of *R.U.R*, however, Čapek concentrates on a sentimental virtue impossible in Unno's fiction by portraying an autonomy that derives from attachment, suffering, and loss. After the extermination of humanity, the Robots force the engineer Alquist to recreate Old Rossum's formula for Robotic life. He rebukes them with the impossibility of his task only to learn that desperation, rather than cool logic, drives the Robots:

Alquist: Robots aren't life! Robots are machines.

Third Robot: We used to be machines, sir; but by means of pain and horror we have become ...

Alquist: Become what?

Second Robot: We have obtained a soul.

Fourth Robot: Something is struggling within us. There are moments when something gets into us. Thoughts come to us that are not our own.⁶²

Alquist later overhears the rare sound of laughter shared between a male Robot named Primus and female Robot modeled after Helena exchanging trifles about dreams and

attractions. Interrupting this scene, Alquist orders Primus to prepare for the dissection of Helena in the hope of discovering the novelty that makes this sentiment possible. Primus refuses and offers himself in her place, a sacrifice that brings Helena to tears. Shocked, Alquist asks, “Oh, child, you can cry? Tell me, what does some Primus matter?”⁶³ Robot Helena answers only by demanding that Primus allow her to be dissected instead. Witnessing this mutual spirit of self-sacrifice, Alquist liberates the two Robots with the injunction “Go, Adam – Go, Eve...” before delivering a message of rebirth which terminates the play: “O nature, nature, life will not perish... It will begin anew with love...”⁶⁴

Contemporary reviewers criticized this finale as “crassly sentimental.”⁶⁵ For Tezuka however, the conclusion of *R.U.R.* may have offered something more than bathos. Indeed, it expresses a dramatic container for a melancholic faith contiguous with the mangaka’s investment in the autonomy of science. The sentiment may ring pathetic, but Primus and Robot Helena are still technological products of scientific knowledge. These artificial humans not only become capable of noble feelings like self-sacrifice but, through the hope vested in their renewal of life itself, literally embody a trust in the limitations of science to produce the conditions of its own autonomy. The same postwar situation of degradation and sterility initiated by technoscientific development conditions the emotion of these technical objects for living toward a future without promise. If, as Tanaka Michio asserts, Tezuka’s encounter with Čapek’s work proved “fortunate” in its provision of a medium to express feelings of “world-weariness” resulting from postwar emotional trauma, it holds equally true that Tezuka stood to

interiorize the play's final image of an autonomous robot sentiment piercing this weariness through its own scientific heritage.⁶⁶

Čapek expressed a version of this melancholic humanism in commenting on his desire to show “what humanity was all about” through *R.U.R.*: “Imagine yourself standing over the grave of mankind; however jaundiced your view, you would surely realize the divine significance of the extinguished species and say – you too: It was a great thing to be a man.”⁶⁷ Like the Robots themselves, Čapek's hypothetical grave acts as a technology of sentiment. It is a rhetorical invitation to feeling calling out to the ensouled, those subordinated to a process of bequeathing to an alien future those things “struggling within us.” The Robots are thus the grave of humanity in a more refined sense: they are the transient remnants of sentiment capable of inducing and perpetuating sentiment itself, the still all-too-human traces of excess, attachment, and vulnerability opened and closed by the scientific spirit that gave rise to them. In presenting this sepulchral image in narrative form, the conclusion of *R.U.R.* dramatizes the transference of a soul from human to organic machine less as a gift of freedom, intellect, or optimism than as the inheritance of a tragic sensitivity to loss, negativity, and failure. It is this sensitivity that recalls and revises the solemn isolation of Unno's technician away from the repulsion of foreign threats and toward the embrace of sentiment itself as a means of moving into the future through the cultivation of a sense that derives from and responds to the failures of science.

The communication of this tragedy through *R.U.R.* furnishes another way to grasp the significance of the dynamic exhibited in Atom's frequent episodes of melancholy, regret, and fear. In early chapters of *Mighty Atom*, an implicit sense of

Atom's emotivity precedes and buttresses an instance of his social isolation as a technological product. The second chapter of the series "Frankenstein," for instance, combines Shelley's outcast creature with the plot of *R.U.R.* to define the structural boundaries against which the sentiment of Atom becomes active. After a corpulent capitalist uses robot labor to automate his robot factory, this robotic self-reproduction comes to generate, as if by accident, a foil to Atom: an "imperfect" robot named Frankenstein who kills his inventors in a rampage. The ensuing moral panic aggravates human-robot relations by encouraging a reactionary discourse about robot subservience, an atmosphere against which the isolated sentimental reflection of Atom becomes distinct. The robot boy quivers and cries as he witnesses the harassment of his surrogate robot father, sinks into a sullen posture after being forced to make a truce with a group of robot revolutionaries, and sits forlorn atop a construction heap soaked in the anonymous electric glare of a cityscape.⁶⁸ Precisely at this moment Atom hears a radio broadcast almost lifted verbatim from Act III of *R.U.R.*, itself a reference to *The Communist Manifesto*, rousing the "Robots of the world!" to revolt.⁶⁹ Atom eventually confronts and thwarts the broadcasters, a gang of humans disguised as robot revolutionaries controlling the supposedly autonomous Frankenstein to foment chaos.

Instead of learning to master various machines in the manner of Unno's technician, here Atom learns to articulate and manage his sentimental education as a relative of Čapek's emotional machines: only because Atom *feels* the absence of sensitivity to sentiment in humans liable to technoscientific excess does he engage in heroic feats. In another early chapter titled "Shootout in the Alps" (1956) a sullen Atom dwells on his inability to understand the beauty of music, a phenomenon which he only perceives as

“a bunch of notes in a special order.”⁷⁰ Despite his friend Ken pointing out that Atom can “get angry and cry sometimes,” Atom pressures Dr. Ochanomizu to modify his body with an artificial heart, a device enabling the feeling of “fear and terror.”⁷¹ Paralyzed by fear in confrontation with a gang of villains at the conclusion of the chapter, Atom’s technical efficiency only kicks in after Ken removes his artificial heart. The final frame pictures Atom promising to stop “being envious of humans and live like a **real robot!**”⁷² And yet Tezuka depicts this forfeiture of deep emotion through a deeper visual irony: in panel that takes up nearly three-quarters of the page, Atom delivers this line oriented toward an aesthetic abundance that he has supposedly rejected, becoming a *rückenfigur* gazing out from a cliff at ragged peaks, billowing clouds, and the bold rays of the rising sun.⁷³ Resonating with the conclusion of *R.U.R.*, the heightened aesthetic profile of this final panel promises a robotic rebirth through the confrontation with and participation in a sublimity of feeling that both exceeds the human and yet remains a distinctly human aesthetic experience.

What defines a “real” robot in this moment, and throughout future chapters of *Mighty Atom*, is an engagement with the melancholic humanism of *R.U.R.*: to be stirred by a sensitivity to loss from a supposedly limited or artificial position to act towards the preservation of sensitivity itself through new attachments. From the vantage of the romantic sublimity that ends “The Shootout in the Alps,” Atom’s decree exposes the redundancy of an “artificial heart” for a robot boy constructed from the autopoietic sentiment of Čapek’s Robots and the lone heroism of Unno’s technician. He merely reaffirms that he is also a creation of science who, in his participation in and reflection of the great foibles of his creators, holds out the possibility of a more-than-human future

based on sensitivity and care. The various blushes, tears, grimaces, and smiles that Atom exhibits in future chapters, as well as his staunch advocacy for robot rights, are, in this sense, not simply random expressions of some supposed human essence that manages to crack through the electro-circuitry of the young robot's brain or an empty gesture toward liberal democracy. Just as *R.U.R.* provides the image of a sentimental robot, a product of science capable of feeling, which entails both utter loss and future attachment, the emotionality of Atom is a technical inheritance from a literary environment that promises renewal of and for science in the face of the violence and devastation caused by science. It is precisely this inheritance which bonds Atom to humanity and which motivates his iconic sentimental heroism.

Conclusion: Mechaphilia and Mech-aesthesis

In a 2002 interview with *Conté* magazine, Go Nagai re-traced the influence of Atom's conflicted stance between violent technical capacity and explicit sensitivity on his own formation as a creator. Setting out from memories of a childhood “almost raised on Atom,” a boy robot who “normally very gentle, in reality hid a great power,” Nagai reveals how he became “captivated by the high degree of [Atom's] perfection as a character, a being with unparalleled power in the form of an angel, as a kind of ideal image.”⁷⁴ The preservation of this ideality underwrites the tactic Nagai used to differentiate his own manga robot in *Mazinger Z* (1972) from existing models. Rather than replicate a robot living to regain his lost humanity in the mold of Atom or the golem *wunderwaffen* of Mitsuteru Yokoyama's *Tetsujin 28* (1956), Nagai externalized the divide between human and robot only to re-internalize and mobilize it. In *Mazinger Z*, humans would pilot, and thus inspire, highly-stylized idealized humanoid machines

not just to battle and destroy, but also to express and preserve those things which were most human in them. Although, as Nagai admitted, the robot-pilot matrix constitutes a technological downgrade when compared to the complex cybernetics of Atom, it nonetheless conveys the “perfection” at the core of Tezuka's robot boy and the questions at the center of the literary heritage he embodies.⁷⁵

This insertion of an impassioned human pilot into the technical complex of a super robot set the stage for the development of the mecha genre that ramified further in the following decades with Yoshiyuki Tomino's *Mobile Suit Gundam* (1979), Hideaki Anno's *Neon Genesis Evangelion* (1998), and remains active in productions like Masami Ôbari's *Brave Bang Bravern!* (2024). The isolation of machinic interaction expressed in *Earth Fortress* and re-articulated through Tezuka here becomes manifest: it is directly of a piece with the emotional reluctance of *Gundam*'s Amuro or *Evangelion*'s Shinji to inhabit their respective humanoid machines of war, to become literally and singularly contained within a sublime artifice resembling and distorting the human mind and body in order to safeguard the sentiments, ideals, and lives from which the pilot becomes temporarily detached. Glimmers of Unno's negotiation between heroic isolation and technoscientific autonomy in this way course throughout the wide range of contemporary popular robots, mechs, and mobile suits that look back to Atom as their indirect forebear.

Yet this genealogy bringing together forlorn feeling and direct confrontation with scientific technologies troubles the notion that scholars such as Thomas Lamarre identify at the center of the mecha genre: mechaphilia, or a “fascination with technoscientific modernity that literally draws boys into militarism, spurring totalitarian forms

of identification with the nation-state.”⁷⁶ Citing Eiji Otsuka, Lamarre dates this mechapilia from the time of Unno's writing during the Pacific War, specifically in its derivation from the introduction of Cartesian ways of thinking and seeing into Japanese military draftsmanship and technology. Through the increasing prominence of psychotechnical motifs such as the “psionic interface” in late-century works like *Evangelion*, however, a narrative tendency arises whereby “Boy pilots become gradually feminized in the sense that operating mecha demand that they be in touch with their feelings and prone to affective communication in a manner previously coded as feminine.”⁷⁷ While certainly true of *Evangelion* and the retinue of series that centered the angst and isolation of their protagonists in its wake, these later emotional reparations of the human and machine division only point more insistently to the complexity of the prior, indeed foundational, entanglement of feeling and force within Atom, the same figure that inspired Nagai and set the genre in motion.

To see the sensitive robot Atom at the root of the mecha genre, then, is to open a space for interpretation that avoids synonymizing fascination with pathologization. Where Lamarre and Otsuka envision a discourse of control or the fantasmatic enjoyment of an isolated Cartesian subject flirting with totalitarianism, Atom's isolated sentiment, his self-mediating struggle with fellow machines, and his violent technical capacity instead represent how, in Jean-Luc Nancy's view, “the Cartesian event... is essentially ambivalent, and the man who is alone in the world is without a doubt also the one who, overtime, cannot but alter the sufficiency of the self...”⁷⁸ From this perspective, the isolation of science and technology indicate a more primary dislocation: “a relation to self” that “supposes a distance to self.”⁷⁹ Unno's instrumentalism and Čapek's

sentimentalism are, in this sense, parallel engagements with the original self-difference that science as a method of abstraction and technology as prosthesis instantiate. In mediating the heroism of the former through the humanism of the latter, Atom embodies this self-difference directly. Unno's self-instrumentalizing technician joins Čapek's posthuman mourner.

If the legacy of Unno's militaristic scientism continues to surface tropes of mechaphilia in Japanese popular culture, as in Leiji's Matsumoto's *Space Battleship Yamato* (1974), where Captain Juzô Okita (explicitly named after Unno) pilots a resurrected WWII submarine through the cosmos, Tezuka's misprision sets up a parallel but contrary dynamic at the beginning of modern anime and manga: a mech-aesthetics or feeling-with machines through the invitation to sense that all machines make available. While shows of force are never far away, Tezuka ultimately figures a character who tempers and negotiates with his self-instrumentalization as an autonomous machine through the inheritance of a sentiment that allows him to communicate with the robots and aliens that populate *Mighty Atom*. In this sensitivity, Atom raises the "question of thinking and experiencing the intensity of a communicating (rather than communicated) sense in accordance with the non-immediacy and heterogeneity of sense, of their 'subjects,' and of their 'channels.'" ⁸⁰ A product of science reckoning with his own limitations and the excesses of other robots, Atom continues Unno's desire to "reorient the direction of scientific development in the world" by embodying the possibility of a scientific spirit based on an opening to sense within the world that technoscientific development supposedly forecloses. ⁸¹

Rather than as a gambit for militaristic power, then, *Mighty Atom* engages the figure of the humanoid machine in a distinctly reflexive way. This little robot, together with his giant descendants, constitute mediated sites of complex identification. Their proliferation across film, toys, video games, and other consumer products simultaneously fulfills Unno's forecast in *Earth Fortress* of a lonely machine future, one defined by mass consumption, technological surveillance, and increasing automation. Yet, like the Robots of *R.U.R.*, these same narratives also defer the realization of this future through the sense and sensitivity purveyed in the very same images of fictional autonomous hi-tech robots. This proliferation of mecha media, like the legacy of Unno's writing coursing through the many human-robot narratives at its core it, remains ambiguous. In it is posed the same conflict captured in the question from Unno engraved on a monument dedicated to his memory in 1964 : "Even as humanity bathes in the blessings of science, it is menaced by dreams of scientific horror. We humans are possessed by this Janus-faced science both blessing and curse, god and devil. Thus, in such a scientific age, should there not be a scientific novel?" Through his invention of Atom, the child robot who would become the father of mecha, Tezuka repeats and revises this question in his own monument to Unno and asks anew: should there not be a scientific manga in which humanity reckons with this contradictory possession, with the blessing and curse uniquely its own?

¹ Natsu Onoda Power, *God of Comics: Osamu Tezuka and the Creation of Post-World War II Manga* (Jackson: University of Mississippi Press, 2009), 170.

² Osamu Tezuka, *Metropolis* (Milwaukie: Dark Horse, 2003), 165.

³ Power, 170.

- ⁴ Kida Jun'ichiro, "Unno Jûza no nikki," in *Unno Jûza kenkyû* 6 (Tokyo: San-ichi Shobo, 1988-1993), 6:5.
- ⁵ See Toshio Ban, *The Osamu Tezuka Story: A Life in Manga and Anime* (Berkeley: Stone Bridge Press, 2016), 132.
- ⁶ Shimamoto Mitsuaki, "Tezuka Osamu wa Unno Jûza wo dou miteita ka," in *Unno Jûza kenkyû* 15 (Tokyo: San-ichi Shobo, 1988-1993), 15:3.
- ⁷ *Ibid.*, 15:4.
- ⁸ Harold Bloom, *The Anxiety of Influence: A Theory of Poetry* (Oxford: Oxford UP, 1997), xxiv.
- ⁹ See Inoue Haruki, "Tezuka Osamu no Tetsuwan Atomu ga kataru gijutsu no daikoku no yukue," *WIRED* (January 1998), 78-83; Nagayama Yasuo, *Nihon SF seishinshi* (Tokyo: Kawade Shobo, 2009), 172-174; Ryan Holmberg, "Osamu Tezuka and the First Story Manga," in *The Mysterious Underground Men* (Brooklyn: PictureBox, 2013), xiv-xvii.
- ¹⁰ Inoue, 80-81.
- ¹¹ *Ibid.*
- ¹² *Ibid.*, 82. See also Inoue Haruki, *Nihon robotto sensouki 1939-1945* (Tokyo: NTT, 2007).
- ¹³ Frederik L. Schodt, *The Astro Boy Essays* (Berkeley: Stone Bridge Press, 2007), 106.
- ¹⁴ Sena Takahiko, *Unno Jûza zenshû*, 5:306.
- ¹⁵ Seth Jacobowitz, "Unno Jûza and the Uses of Science in Prewar Japanese Popular Fiction," in *New Directions in Popular Fiction* (London: Palgrave, 2016), 159.
- ¹⁶ Miri Nakamura, "Marking bodily differences: mechanized bodies in Hirabayashi Hatsunosuke's 'Robot' and early Showa robot literature," *Japan Forum* 19:2 (2007), 179.
- ¹⁷ James Reichert, "Deviance and Social Darwinism in Edogawa Ranpô's erotic-grotesque thriller *Kotô no oni*," *Journal of Japanese Studies* 27:1 (2001), 114.
- ¹⁸ Unno Jûza, "Jinzô monogatari," in *Unno Jûza zenshû*, 14:291.
- ¹⁹ *Ibid.*, 295.
- ²⁰ Unno Jûza, "Jinzô ningen jiken," in *Unno Jûza zenshû*, 5:38.
- ²¹ *Ibid.*, 39.
- ²² *Ibid.*, 49.
- ²³ Unno, "Jinzô monogatari," 14:299.
- ²⁴ Jacobowitz, 159.
- ²⁵ Itagaki, Takao, *Kikai to geijutsu to no kôryô*, Iwanami Shoten, 1929.
- ²⁶ Unno Jûza, "Eighteen O'clock Music Bath," *Fast Forward Japan*, trans. JD Wisgo (Arigatai Books, 2021), 142.
- ²⁷ *Ibid.*, 143.
- ²⁸ *Ibid.*, 149-150.
- ²⁹ *Ibid.*, 161.
- ³⁰ For more on the notable congruence between the multifunctional boy robot Mr. F and Atom, see Shiozaki Noboru, "Tetsuwan Atomu no rûtsu wo saguru," in *Tanjô! Tezuka Osamu: manga no kamisama wo sodateta bakkuguraundo* (Tokyo: Asahi Sonorama, 1998), 163-166.
- ³¹ Unno Jûza, "Jinzô ningen Efu shi," in *Unno Jûza zenshû*, 6:291.
- ³² Sari Kawana, *The Uses of Literature in Modern Japan: Histories and Cultures of the Book* (New York: Bloomsbury, 2018), 53.
- ³³ *Ibid.*, 65.
- ³⁴ *Ibid.*, 66.
- ³⁵ Osamu Tezuka, *Tezuka Osamu taidanshû* (Tokyo: Kodansha, 1997), 3:53-54.
- ³⁶ Ran Zwigenberg, *Nuclear Minds* (Chicago: Chicago University Press, 2023), 37.
- ³⁷ Unno Jûza, *Chikyû yôsai*, in *Unno Jûza zenshû*, 7:22.
- ³⁸ *Ibid.*, 22-23.
- ³⁹ *Ibid.*, 23.

- ⁴⁰ Ibid., 41.
⁴¹ Ibid., 50.
⁴² Tezuka, *Tezuka Osamu taidanshû*, 54.
⁴³ Tezuka, *Doctor Mars* (Tokyo: Kodansha, 1994), 74, 83.
⁴⁴ Takeuchi Osamu, *Tezuka Osamu ron* (Tokyo: Heibonsha, 1992), 48.
⁴⁵ Tezuka, *Doctor Mars*, 109;77.
⁴⁶ Ibid., 26.
⁴⁷ Ibid., 27.
⁴⁸ Iwamoto Yûichirô, “Robotto no hyoushou,” *Toyama Hikaku Bunkaku* 9 (2016), 147.
⁴⁹ Osamu Tezuka, *Astro Boy*, trans. Frederik L. Schodt (Milwaukie: Dark Horse, 2015-2017), 3:14.
⁵⁰ Ibid., 18.
⁵¹ Alicia Gibson, “Out of Death, An Atomic Consecration to Life: *Astro Boy* and Hiroshima’s Long Shadow,” *Mechademia*, ed. Frenchy Lunning, 8 (Minneapolis: University of Minnesota Press, 2013), 317.
⁵² Tezuka, *Astro Boy*, 1:26.
⁵³ Tezuka, *Astro Boy*: 1:28-29.
⁵⁴ Gibson, 315.
⁵⁵ Ban, 314.
⁵⁶ Otsuka Eiji, *Atomu no meidai: Tezuka Osamu to sengo manga no shudai* (Tokyo: Kadokawa, 2009), 178; Komatsu Sakyo, “Interview with Komatsu Sakyo,” *The Robot Chronicles* (Tokyo: Asahi Shinbun, 2002), 65.
⁵⁷ Karel Čapek, *R.U.R.* (New York: Penguin, 2004), 56.
⁵⁸ Kara Reilly, *Automata and Mimesis on the Stage of Theatre History* (London: Palgrave, 2011), 149.
⁵⁹ Ibid.
⁶⁰ Karel Čapek, “The Meaning of *R.U.R.*” *Saturday Review* 136 (23 July 1923), 79.
⁶¹ Jacobowitz, 159.
⁶² Čapek, *R.U.R.*, 75.
⁶³ Ibid., 83.
⁶⁴ Ibid., 84.
⁶⁵ Reilly, 171.
⁶⁶ Tanaka Michio, *Atomu wa boku ga koroshimashita* (Tokyo: Sunmark, 2002), 144.
⁶⁷ Čapek, “The Meaning of *R.U.R.*”
⁶⁸ Tezuka, *Astro Boy*, 7:257; 271; 277.
⁶⁹ Tezuka, *Astro Boy*, 7:278.
⁷⁰ Tezuka, *Astro Boy*, 4:639.
⁷¹ Ibid., 630-640.
⁷² Ibid., 667.
⁷³ Tezuka, *Astro Boy*, 4:667.
⁷⁴ Nagai, 18-19.
⁷⁵ Ibid., 19.
⁷⁶ Thomas Lamarre, *The Anime Machine* (Minneapolis: Minnesota University Press, 2009), 212.
⁷⁷ Ibid., 216.
⁷⁸ Jean-Luc Nancy, *The Fragile Skin of the World* (Cambridge: Polity, 2021), 29.
⁷⁹ Ibid., 30.
⁸⁰ Ibid., 37.
⁸¹ Sari Kawana, *Murder Most Modern: Detective Fiction and Japanese Culture* (Minneapolis: University of Minnesota Press, 2008), 163.

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